Bench	Task	Trials	Raw	+Workflow	+Skills	Raw Tok	WF Tok	SK Tok	ΔP_{SK-Raw}	ρ_{SK-Raw}
Terminal-Bench-2										
TB2	crack-7z-hash	5	80.0%	0.0%	20.0%	986K	332K	683K	-60.0%	-30.8%
TB2	overfull-hbox	5	80.0%	40.0%	20.0%	555K	350K	723K	-60.0%	+30.3%
TB2	financial-document-processor	5	60.0%	0.0%	0.0%	1.1M	147K	647K	-60.0%	-41.0%
TB2	model-extraction-relu-logits	5	60.0%	40.0%	0.0%	482K	198K	191K	-60.0%	-60.3%
TB2	query-optimize	5	40.0%	0.0%	0.0%	642K	297K	875K	-40.0%	+36.2%
TB2	torch-tensor-parallelism	5	40.0%	0.0%	0.0%	174K	172K	176K	-40.0%	+0.8%
TB2	fix-code-vulnerability	5	80.0%	60.0%	80.0%	1.4M	511K	642K	+0.0%	-54.5%
TB2	path-tracing	5	80.0%	0.0%	80.0%	2.2M	247K	2.3M	+0.0%	+5.5%
TB2	path-tracing-reverse	5	80.0%	0.0%	80.0%	4.0M	855K	4.2M	+0.0%	+5.3%
TB2	regex-chess	5	80.0%	20.0%	80.0%	3.0M	420K	1.7M	+0.0%	-44.9%
TB2	tune-mjcf	5	20.0%	20.0%	20.0%	453K	143K	436K	+0.0%	-3.8%
TB2	build-pov-ray	5	80.0%	60.0%	100.0%	1.4M	771K	2.0M	+20.0%	+49.9%
TB2	chess-best-move	5	80.0%	60.0%	100.0%	361K	99K	318K	+20.0%	-11.8%
TB2	large-scale-text-editing	5	80.0%	20.0%	100.0%	403K	267K	392K	+20.0%	-2.6%
TB2	log-summary-date-ranges	5	80.0%	40.0%	100.0%	163K	56K	115K	+20.0%	-29.5%
TB2	mcmc-sampling-stan	5	80.0%	20.0%	100.0%	10.8M	444K	5.9M	+20.0%	-45.2%
TB2	portfolio-optimization	5	80.0%	80.0%	100.0%	391K	328K	368K	+20.0%	-5.7%
TB2	pytorch-model-recovery	5	40.0%	40.0%	60.0%	794K	193K	1.1M	+20.0%	+37.6%
TB2	rstan-to-pystan	5	80.0%	0.0%	100.0%	4.1M	923K	3.6M	+20.0%	-11.5%
TB2	sanitize-git-repo	5	80.0%	100.0%	100.0%	2.2M	831K	829K	+20.0%	-62.7%
TB2	winning-avg-corewars	5	80.0%	40.0%	100.0%	4.7M	445K	3.3M	+20.0%	-28.9%
TB2	hf-model-inference	5	60.0%	0.0%	80.0%	1.1M	397K	802K	+20.0%	-28.4%
TB2	adaptive-rejection-sampler	5	20.0%	20.0%	60.0%	1.7M	445K	1.1M	+40.0%	-31.2%
TB2	dna-insert	5	20.0%	0.0%	60.0%	454K	187K	359K	+40.0%	-20.9%
TB2	break-filter-js-from-html	5	40.0%	0.0%	80.0%	376K	142K	156K	+40.0%	-58.6%
TB2	mteb-leaderboard	5	60.0%	0.0%	100.0%	19.8M	1.1M	2.7M	+40.0%	-86.6%
TB2	write-compressor	5	60.0%	20.0%	100.0%	248K	94K	723K	+40.0%	+191.3%
TB2	cancel-async-tasks	5	40.0%	20.0%	100.0%	69K	46K	133K	+60.0%	+93.1%
TB2	extract-elf	5	40.0%	60.0%	100.0%	722K	155K	186K	+60.0%	-74.3%
TB2	qemu-startup	5	20.0%	80.0%	80.0%	758K	199K	174K	+60.0%	-77.1%
TB2	pypi-server	5	20.0%	0.0%	100.0%	499K	315K	296K	+80.0%	-40.7%
SkillsBench										
SB	weighted-gdp-calc	5	80.0%	40.0%	0.0%	633K	457K	525K	-80.0%	-17.0%
SB	protein-expression-analysis	5	80.0%	80.0%	40.0%	529K	454K	560K	-40.0%	+5.9%
SB	virtualhome-agent-planning	5	40.0%	100.0%	0.0%	284K	186K	151K	-40.0%	-46.8%
SB	data-to-d3	5	60.0%	40.0%	20.0%	882K	918K	850K	-40.0%	-3.7%
SB	energy-market-pricing	5	60.0%	20.0%	20.0%	866K	880K	1.4M	-40.0%	+58.8%
SB	hvac-control	5	80.0%	60.0%	60.0%	220K	363K	299K	-20.0%	+35.9%
SB	adaptive-cruise-control	5	20.0%	60.0%	0.0%	5.7M	3.2M	3.3M	-20.0%	-42.8%
SB	paper-anonymizer	5	40.0%	40.0%	20.0%	940K	962K	507K	-20.0%	-46.0%
SB	r2r-mpc-control	5	60.0%	60.0%	60.0%	526K	397K	391K	+0.0%	-25.7%
SB	dapt-intrusion-detection	5	80.0%	100.0%	100.0%	277K	420K	322K	+20.0%	+16.2%
SB	earthquake-phase-association	5	80.0%	80.0%	100.0%	2.1M	1.6M	1.5M	+20.0%	-30.4%
SB	glm-lake-mendota	5	80.0%	80.0%	100.0%	2.0M	2.0M	1.7M	+20.0%	-14.5%
SB	grid-dispatch-operator	5	40.0%	20.0%	60.0%	635K	1.2M	334K	+20.0%	-47.3%
SB	video-filler-word-remover	5	40.0%	20.0%	60.0%	2.8M	3.0M	771K	+20.0%	-72.2%
SB	fix-druid-loophole-cve	5	20.0%	20.0%	40.0%	13.4M	7.6M	15.5M	+20.0%	+15.7%
SB	organize-messy-files	5	20.0%	40.0%	40.0%	602K	637K	955K	+20.0%	+58.6%
SB	trend-anomaly-causal-inference	5	20.0%	0.0%	60.0%	1.9M	5.3M	1.9M	+40.0%	-1.5%
SB	fix-erlang-ssh-cve	5	60.0%	80.0%	100.0%	2.3M	1.8M	1.5M	+40.0%	-34.0%
SB	fix-visual-stability	5	60.0%	100.0%	100.0%	4.9M	3.1M	4.4M	+40.0%	-10.4%
SB	invoice-fraud-detection	5	60.0%	0.0%	100.0%	359K	229K	237K	+40.0%	-34.1%
SB	dialogue-parser	5	40.0%	20.0%	100.0%	488K	516K	525K	+60.0%	+7.6%
SB	financial-modeling-qa	5	40.0%	40.0%	100.0%	827K	728K	338K	+60.0%	-59.1%
SB	react-performance-debugging	5	40.0%	60.0%	100.0%	3.8M	4.4M	3.6M	+60.0%	-6.3%
SB	gravitational-wave-detection	5	20.0%	80.0%	100.0%	2.5M	1.4M	761K	+80.0%	-70.1%
Terminal-Bench-Pro										
TB-Pro	fix-gomoku-win-detection	5	80.0%	100.0%	0.0%	92K	82K	116K	-80.0%	+26.0%
TB-Pro	decode-go-ctf-credentials	5	80.0%	80.0%	20.0%	958K	786K	606K	-60.0%	-36.8%
TB-Pro	implement-nonogram-puzzle-solver	5	60.0%	100.0%	0.0%	130K	325K	224K	-60.0%	+72.1%
TB-Pro	mongodb-sales-aggregation-engine	5	60.0%	60.0%	0.0%	167K	156K	124K	-60.0%	-25.8%
TB-Pro	python-jacobi-eigen-solver-conversion	5	60.0%	0.0%	0.0%	198K	209K	226K	-60.0%	+14.2%
TB-Pro	train-loan-default-logreg-model	5	80.0%	100.0%	60.0%	147K	116K	127K	-20.0%	-13.3%
TB-Pro	jq-github-contributor-report	5	40.0%	80.0%	20.0%	111K	132K	266K	-20.0%	+140.8%
TB-Pro	portfolio-risk-analyzer-script	5	20.0%	20.0%	0.0%	182K	151K	210K	-20.0%	+15.2%
TB-Pro	recover-and-sanitize-git-db-credential	5	20.0%	0.0%	0.0%	548K	327K	382K	-20.0%	-30.3%
TB-Pro	flood-escape-path-planner	5	60.0%	60.0%	40.0%	214K	196K	262K	-20.0%	+22.4%
TB-Pro	build-uci-chess-engine-to-beat-multiple-bots	5	40.0%	40.0%	40.0%	1.1M	2.1M	664K	+0.0%	-40.0%
TB-Pro	fix-xor-neural-network-instability	5	80.0%	80.0%	80.0%	544K	777K	767K	+0.0%	+41.0%
TB-Pro	implement-chemical-equilibrium-solver	5	20.0%	0.0%	20.0%	215K	319K	522K	+0.0%	+142.5%
TB-Pro	implement-depgraph-dependency-resolver	5	20.0%	40.0%	20.0%	468K	222K	227K	+0.0%	-51.4%
TB-Pro	improve-code-similarity-feature-extraction	5	80.0%	60.0%	80.0%	803K	914K	478K	+0.0%	-40.5%
TB-Pro	optimize-urdf-robot-for-pybullet	5	60.0%	60.0%	60.0%	1.6M	1.8M	2.2M	+0.0%	+35.4%
TB-Pro	repair-broken-shell-data-pipeline	5	60.0%	20.0%	60.0%	759K	903K	625K	+0.0%	-17.6%
TB-Pro	train-fraud-detection-model	5	40.0%	0.0%	40.0%	180K	183K	147K	+0.0%	-18.8%
TB-Pro	regex-bitcoin-p2pkh-extraction	5	40.0%	100.0%	60.0%	65K	42K	133K	+20.0%	+104.7%
TB-Pro	reverse-engineer-kvstore-binary-protocol	5	80.0%	80.0%	100.0%	810K	402K	1.2M	+20.0%	+46.6%
TB-Pro	secure-postgres-schema-migration	5	80.0%	100.0%	100.0%	634K	415K	627K	+20.0%	-1.1%
TB-Pro	advanced-json-to-rfc4180-csv-converter	5	60.0%	40.0%	80.0%	401K	635K	439K	+20.0%	+9.6%
TB-Pro	validate-and-solve-sudoku	5	60.0%	100.0%	80.0%	438K	377K	264K	+20.0%	-39.8%
TB-Pro	compare-cmake-build-types-performance	5	60.0%	100.0%	100.0%	111K	98K	170K	+40.0%	+53.2%
TB-Pro	normalize-invoice-pdfs-to-csv	5	40.0%	40.0%	80.0%	212K	117K	230K	+40.0%	+8.2%
TB-Pro	present-known-plaintext-key-recovery	5	60.0%	100.0%	100.0%	5.9M	1.2M	1.8M	+40.0%	-68.9%
TB-Pro	python-sql-injection-sanitizer	5	40.0%	100.0%	80.0%	201K	202K	181K	+40.0%	-10.0%
TB-Pro	setup-mlflow-sqlite-server	5	60.0%	20.0%	100.0%	1.0M	901K	266K	+40.0%	-74.2%
TB-Pro	text-image-ocr-pipeline	5	60.0%	60.0%	100.0%	109K	122K	128K	+40.0%	+17.8%
TB-Pro	train-matrix-factorization-embeddings	5	60.0%	20.0%	100.0%	351K	172K	360K	+40.0%	+2.5%
TB-Pro	train-sarsa-taxi-agent	5	60.0%	60.0%	100.0%	422K	818K	264K	+40.0%	-37.4%
TB-Pro	implement-tensor-parallel-matmul	5	40.0%	100.0%	100.0%	165K	509K	209K	+60.0%	+26.8%
TB-Pro	sokoban-bfs-move-with-fallback	5	20.0%	100.0%	100.0%	111K	193K	191K	+80.0%	+72.0%

bottomrule